

LD<sub>50</sub> (rat, IV): 5.5 g/kg  
 LD<sub>50</sub> (rat, oral): 5.8 g/kg

## 15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of material handled. Acetone is a skin and eye irritant (*see* Section 14); therefore gloves, eye protection and a respirator are recommended. In the UK, the long-term (8-hour TWA) workplace exposure limit for acetone is 1210 mg/m<sup>3</sup> (500 ppm). The short-term (15-minute) exposure limit is 3620 mg/m<sup>3</sup> (1500 ppm).<sup>(6)</sup>

## 16 Regulatory Status

Included in the FDA Inactive Ingredients Database (inhalation solution; oral tablets; topical preparations). Included in the Canadian List of Acceptable Non-medicinal Ingredients. Included in nonparenteral medicines licensed in the UK.

## 17 Related Substances

## 18 Comments

A specification for acetone is included in the *Japanese Pharmaceutical Excipients* (JPE).<sup>(7)</sup>

The EINECS number for acetone is 200-662-2. The PubChem Compound ID (CID) for acetone is 180.

## 19 Specific References

- 1 Ash M, Ash I. *Handbook of Pharmaceutical Additives*, 3rd edn. Endicott, NY: Synapse Information Resources, 2007; 430.
- 2 Tang ZG *et al.* Surface properties and biocompatibility of solvent-cast poly[ε-caprolactone] films. *Biomaterials* 2004; 25(19): 4741–4748.
- 3 Ruan G, Feng SS. Preparation and characterization of poly(lactic acid)–poly(ethylene glycol)–poly(lactic acid) (PLA-PEG-PLA) microspheres for controlled release of paclitaxel. *Biomaterials* 2003; 24(27): 5037–5044.
- 4 Todd RG, Wade A, eds. *The Pharmaceutical Codex*, 11th edn. London: Pharmaceutical Press, 1979; 6.
- 5 Lewis RJ, ed. *Sax's Dangerous Properties of Industrial Materials*, 11th edn. New York: Wiley, 2004; 22–23.
- 6 Health and Safety Executive. EH40/2005: *Workplace Exposure Limits*. Sudbury: HSE Books, 2005 (updated 2007). <http://www.hse.gov.uk/coshh/table1.pdf> (accessed 5 February 2009).
- 7 Japan Pharmaceutical Excipients Council. *Japanese Pharmaceutical Excipients* 2004. Tokyo: Yakuji Nippo, 2004; 35–36.

## 20 General References

## 21 Author

AH Kibbe.

## 22 Date of Revision

5 February 2009.

# Acetyltributyl Citrate

## 1 Nonproprietary Names

USP-NF: Acetyltributyl Citrate

PhEur: Tributyl Acetylcitrate

## 2 Synonyms

Acetylbutyl citrate; acetylcitric acid, tributyl ester; ATBC; *Citroflex A-4*; tributyl acetylcitrate; tributylis acetylcitras; tributyl O-acetylcitrate; tributyl citrate acetate.

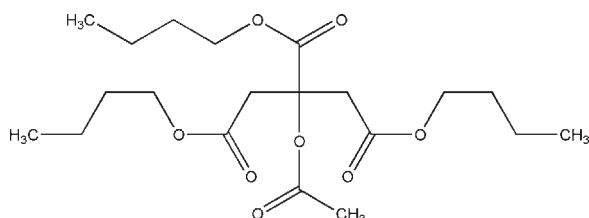
## 3 Chemical Name and CAS Registry Number

1,2,3-Propanetricarboxylic acid, 2-acetyloxy, tributyl ester [77-90-7]

## 4 Empirical Formula and Molecular Weight

C<sub>20</sub>H<sub>34</sub>O<sub>8</sub> 402.5

## 5 Structural Formula



## 6 Functional Category

Plasticizer.

## 7 Applications in Pharmaceutical Formulation or Technology

Acetyltributyl citrate is used to plasticize polymers in formulated pharmaceutical coatings,<sup>(1–5)</sup> including capsules, tablets, beads, and granules for taste masking, immediate release, sustained-release and enteric formulations.

## 8 Description

Acetyltributyl citrate is a clear, odorless, practically colorless, oily liquid.

## 9 Pharmacopeial Specifications

*See* Table I.

## 10 Typical Properties

*Acid value* 0.02

*Boiling point* 326°C (decomposes)

*Flash point* 204°C

*Pour point* –59°C

*Solubility* Miscible with acetone, ethanol, and vegetable oil; practically insoluble in water.

*Viscosity (dynamic)* 33 mPa s (33 cP) at 25°C

**Table I:** Pharmacopeial specifications for acetyltributyl citrate.

| Test                    | PhEur 6.3   | USP32-NF27    |
|-------------------------|-------------|---------------|
| Identification          | +           | +             |
| Appearance              | +           | —             |
| Characters              | +           | —             |
| Specific gravity        | —           | 1.045–1.055   |
| Refractive index        | 1.442–1.445 | 1.4410–1.4425 |
| Sulfated ash            | ≤0.1%       | —             |
| Acidity                 | +           | +             |
| Water                   | ≤0.25%      | ≤0.25%        |
| Heavy metals            | ≤10 ppm     | ≤0.001%       |
| Related substances      | +           | —             |
| Assay (anhydrous basis) | 99.0–101.0% | ≥99.0%        |

## 11 Stability and Storage Conditions

Acetyltributyl citrate should be stored in a well-closed container in a cool, dry location at temperatures not exceeding 38°C. When stored in accordance with these conditions, acetyltributyl citrate is a stable product.

## 12 Incompatibilities

Acetyltributyl citrate is incompatible with strong alkalis and oxidizing materials.

## 13 Method of Manufacture

Acetyltributyl citrate is prepared by the esterification of citric acid with butanol followed by acylation with acetic anhydride.

## 14 Safety

Acetyltributyl citrate is used in oral pharmaceutical formulations and films intended for direct food contact. It is also used in self-adhesive thin films used for topical delivery systems.<sup>(6)</sup> It is generally regarded as a relatively nontoxic and nonirritating material. However, ingestion of large quantities may be harmful.

LD<sub>50</sub> (cat, oral): >50 mL/kg<sup>(7)</sup>

LD<sub>50</sub> (mouse, IP): >4 g/kg

LD<sub>50</sub> (rat, oral): >31.5 g/kg

## 15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of material handled. Acetyltributyl citrate is slightly irritating to the eyes and may be irritating to the respiratory system as a mist or at elevated temperatures. Gloves and eye protection are recommended for normal handling, and a respirator is recommended when using acetyltributyl citrate at elevated temperatures.

## 16 Regulatory Status

Included in FDA Inactive Ingredients Database (oral capsules and tablets). Included in nonparenteral medicines licensed in the UK. Approved in the USA for direct food contact in food films.

## 17 Related Substances

Acetyltriethyl citrate; tributyl citrate; triethyl citrate.

## 18 Comments

Acetyltributyl citrate is used as a plasticizer in food contact films, although it has been known to migrate from food-grade PVC films into high-fat foods such as olive oil.<sup>(8)</sup>

Poly lactide plasticized with acetyltributyl citrate has been investigated as a biodegradable barrier for use in guided-tissue regeneration therapy.<sup>(9)</sup>

The EINECS number for acetyltributyl citrate is 201-067-0. The PubChem Compound ID (CID) for acetyltributyl citrate is 6505.

## 19 Specific References

- Gutierrez-Rocca JC, McGinity JW. Influence of water soluble and insoluble plasticizer on the physical and mechanical properties of acrylic resin copolymers. *Int J Pharm* 1994; 103: 293–301.
- Lehmann K. Chemistry and application properties of polymethacrylate coating systems. McGinity JW, ed. *Aqueous Polymeric Coatings for Pharmaceutical Dosage Forms*. New York: Marcel Dekker, 1989; 153–245.
- Steuernagel CR. Latex emulsions for controlled drug delivery. McGinity JW, ed. *Aqueous Polymeric Coatings for Pharmaceutical Dosage Forms*. New York: Marcel Dekker, 1989; 1–61.
- Gutierrez-Rocca JC, McGinity JW. Influence of aging on the physical-mechanical properties of acrylic resin films cast from aqueous dispersions and organic solutions. *Drug Dev Ind Pharm* 1993; 19(3): 315–332.
- Repka MA *et al.* Influence of plasticisers and drugs on the physical-mechanical properties of hydroxypropylcellulose films prepared by hot melt extrusion. *Drug Dev Ind Pharm* 1999; 25(5): 625–633.
- Lieb S *et al.* Self-adhesive thin films for topical delivery of 5-aminolevulinic acid. *Eur J Pharm Biopharm* 2002; 53(1): 99–106.
- Lewis RJ, ed. *Sax's Dangerous Properties of Industrial Materials*, 11th edn. New York: Wiley, 2004; 3512.
- Goulas AE *et al.* Effect of high-dose electron beam irradiation on the migration of DOA and ATBC plasticizers from food-grade PVC and PVDC/PVC films, respectively, into olive oil. *J Food Prot* 1998; 61(6): 720–724.
- Dorfer CE *et al.* Regenerative periodontal surgery in interproximal intrabony defects with biodegradable barriers. *J Clin Periodontol* 2000; 27(3): 162–168.

## 20 General References

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## 21 Authors

ME Quinn, PJ Sheskey.

## 22 Date of Revision

18 February 2009.



# Acetyltriethyl Citrate

## 1 Nonproprietary Names

USP-NF: Acetyltriethyl Citrate

## 2 Synonyms

ATEC; *Citroflex* A-2; triethyl acetylcitrate; triethyl O-acetylcitrate; triethyl citrate acetate.

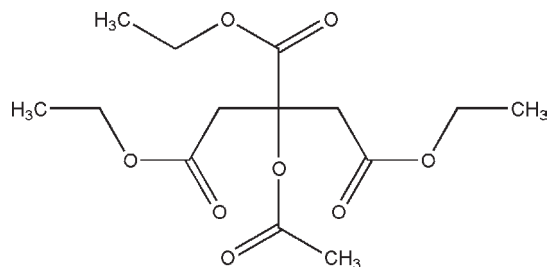
## 3 Chemical Name and CAS Registry Number

1,2,3-Propanetricarboxylic acid, 2-acetyloxy, triethyl ester [77-89-4]

## 4 Empirical Formula and Molecular Weight

C<sub>14</sub>H<sub>22</sub>O<sub>8</sub> 318.3

## 5 Structural Formula



## 6 Functional Category

Plasticizer.

## 7 Applications in Pharmaceutical Formulation or Technology

Acetyltriethyl citrate is used to plasticize polymers in formulated pharmaceutical coatings.<sup>(1)</sup> The coating applications include capsules, tablets, beads and granules for taste masking, immediate release, sustained-release and enteric formulations.<sup>(2-5)</sup> It is also used in diffusion-controlled release drug delivery systems.<sup>(6)</sup>

## 8 Description

Acetyltriethyl citrate occurs as a clear, odorless, practically colorless oily liquid.

## 9 Pharmacopeial Specifications

See Table I.

**Table I:** Pharmacopeial specifications for acetyltriethyl citrate.

| Test                    | USP32-NF27  |
|-------------------------|-------------|
| Identification          | +           |
| Specific gravity        | 1.135–1.139 |
| Refractive index        | 1.432–1.441 |
| Acidity                 | +           |
| Water                   | ≤0.3%       |
| Heavy metals            | ≤0.001%     |
| Assay (anhydrous basis) | ≥99.0%      |

## 10 Typical Properties

*Acid value* 0.02

*Boiling point* 294°C (decomposes)

*Flash point* 188°C

*Pour point* –43°C

*Solubility* Soluble 1 in 140 of water; miscible with acetone, ethanol, and propan-2-ol.

*Viscosity (dynamic)* 54 mPa s (54 cP) at 25°C.

## 11 Stability and Storage Conditions

Acetyltriethyl citrate should be stored in dry, closed containers at temperatures not exceeding 38°C. When stored in accordance with these conditions, acetyltriethyl citrate is a stable product.

## 12 Incompatibilities

Acetyltriethyl citrate is incompatible with strong alkalis and oxidizing materials.

## 13 Method of Manufacture

Acetyltriethyl citrate is prepared by the esterification of citric acid with ethanol followed by acylation with acetic anhydride.

## 14 Safety

Acetyltriethyl citrate is used in oral pharmaceutical formulations and is generally regarded as a nontoxic and nonirritating material. However, ingestion of large quantities may be harmful.

LD<sub>50</sub> (cat, oral): 8.5 g/kg<sup>(7)</sup>

LD<sub>50</sub> (mouse, IP): 1.15 g/kg

LD<sub>50</sub> (rat, oral): 7 g/kg

## 15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of material handled. Acetyltriethyl citrate may be irritating to the eyes or the respiratory system as a mist or at elevated temperatures. Gloves and eye protection are recommended for normal handling and a respirator is recommended if used at elevated temperatures.

## 16 Regulatory Status

Approved in the USA for direct food contact in food films.

## 17 Related Substances

Acetyltributyl citrate; tributyl citrate; triethyl citrate.

## 18 Comments

The EINECS number for acetyltriethyl citrate is 201-066-5. The PubChem Compound ID (CID) for acetyltriethyl citrate is 6504.

## 19 Specific References

- Jensen JL *et al.* Variables that affect the mechanism of drug release from osmotic pumps coated with acrylate/methacrylate copolymer latexes. *J Pharm Sci* 1995; 84: 530–533.
- Gutierrez-Rocca JC, McGinity JW. Influence of water soluble and insoluble plasticizer on the physical and mechanical properties of acrylic resin copolymers. *Int J Pharm* 1994; 103: 293–301.
- Lehmann K. Chemistry and application properties of polymethacrylate coating systems. McGinity JW, ed. *Aqueous Polymeric Coatings for*